

Pixelated Image Abstraction via Power Diagram

Tao Li, Yuyou Yao, Wenming Wu, and Liping Zheng*

School of Computer Science and Information Engineering, Hefei University of
Technology, Hefei, China
zhenglip@hfut.edu.cn

Abstract. Image abstraction is increasingly used in the field of computer graphics and computer vision. Existing methods divide the input images into disjoint regions and generate the abstract images with a reduced color palette. However, the generated images are not satisfactory due to the presence of blurred and unsmooth boundaries, etc. To this end, a pixelated image abstraction method via power diagrams is introduced in this paper to generate high-quality abstract images. Firstly, a pre-processing step is used to capture the significant information in the input image, which is divided into core and non-core parts according to the importance of each pixel. Then, a density map is calculated on basis of the division and grayscale value of each pixel. Furthermore, the primal image is partitioned into polygonal subregions with equal area constraint based on power diagrams. Based on the average RGB color of the 3×3 grids centered at its centroid, the pixels in each polygonal subregion are colored to produce the output abstract image. Experimental results demonstrate that the proposed method could generate high-quality images with abstraction controllable and smooth boundaries.

Keywords: Image abstraction · Segmentation · Pixelated · Power diagram.

1 Introduction

Pixel images are widely used in our daily life, i.e., the smartphones and computer monitors utilize pixel art to convey information to users [13]. With the development of software and hardware in the photograph technology, an increasing number of high-quality images has emerged, which usually contain colorful pixels. Despite these images with colorful pixels bring good experience to users, the problem of data storage and image processing also arises [8]. Hence, it is important to simplify images while preserving the significant pixels.

Image abstraction [18] is an efficient technique for simplifying images by generating a simpler image with less memory consumption and fewer colorful pixels from an input image. Existing image abstraction methods simplify images by considering two aspects: the structure and the color of an image. The former uses less structure to represent an image, while the latter uses less color. On the basis of this, existing image abstraction methods could be broadly classified into

two categories: segmentation based image abstraction method [1, 10, 9, 14] and filter based image abstraction method [22, 16, 15, 21].

Segmentation based image abstraction method divides the primal image into disjoint subregions, each of which is further filled with the same color [1]. DeCarlo et al. [7] introduce a visual perception based non-uniform image abstraction algorithm, which converts an input image into a line-drawing style consisting of multi-regions with same color. High-attention regions are preserved to remain more detailed information, while these regions with low-attention are simplify filled. However, the input image is segmented on basis of the visual perception, which requires an expensive eye tracker and is inefficient. Ren et al. [19] propose the concept of superpixels to divide the input image into irregular blocks, each of which is composed of these pixels with similar features. The superpixels greatly reduce the complex of images, and generate satisfactory abstract images. Several researches [1] have studies the superpixels and present some algorithms for image abstraction [10, 9]. Nevertheless, the image abstraction method based on superpixels may generate low-quality abstract images with unsmooth boundaries (the boundaries between significant and non-significant pixels in the following).

Different from the segmentation based method, the filter based image abstraction method use the Mean-Shift filter to solve the image abstraction problem in video [22]. Specifically, Wang et al. [22] develop an anisotropic Mean-Shift filter to segment video data into continuous data, thus solving the abstraction problem of offline video to achieve good spatio-temporal correlation [23]. However, the smoothness and accuracy of the segmentation are still the challenges, and the large amount of time consumption is also a problem to be solved with the filter based image abstraction method.

Therefore, we focus on the limitations of previous work, namely the unsmooth boundaries and the uncontrollability of generated abstract images. In this paper, we proposed a novel method for pixelated image abstraction with power diagrams. A pre-processing step is required to capture these pixels with important information in the input image, which is divided into core (with significant pixels) and non-core (without significant pixels) parts. Then, a density map is initialized based on the division and grayscale value of each pixel in the input image. Additionally, the primal image is partitioned into polygonal subregions with the same area constraint based on the power diagram, each of which is filled with the average RGB color of the 3×3 grids centered on its centroid. Benefiting from the properties of polygonal subregions, our method could generate high-quality abstract images with smooth boundaries. Besides, the number of sites in the power diagram based image segmentation and the initialized density map give our method with the ability of generating controllable abstract images. The main contribution of our work is as follows:

- We introduce a pixelated image abstraction method via power diagrams to generate high-quality abstract images with a controllable abstraction degree.
- An image segmentation method based on the power diagram to produce polygonal subregions (hexagon is optimal), which makes the generated abstract images with smooth boundaries.

The remainder of this paper is organized as follows. Section 2 briefly reviews the concepts of power diagrams and the image abstraction methods. Our method is introduced in Section 3. Some experimental results are provided in Section 4, and some conclusions are given in Section 5.

2 Related work

In this section, we introduce the concepts of the power diagram and the centroidal capacity constraint power diagram (CCCPD). Besides, a briefly review of image abstraction methods is provided.

2.1 Power diagram

Given the domain $\Omega \subset E^d$ and a set of sites $\mathbf{S} = \{\mathbf{s}_i\}_{i=1}^n$ (also called “points” or “generators”), the Voronoi diagram defines a partition, dividing the domain Ω into n disjoint subregions $V(\mathbf{S}) = \{V(\mathbf{s}_i)\}_{i=1}^n$. Each subregion (also called “Voronoi cell”) $V(\mathbf{s}_i)$ of the site $\mathbf{s}_i$ is defined:

$$V(\mathbf{s}_i) = \{\mathbf{s} \mid \|\mathbf{s} - \mathbf{s}_i\| \leq \|\mathbf{s} - \mathbf{s}_j\|, \forall j \neq i\} \quad (1)$$

where $\|\cdot\|$ is the Euclidean distance.

As an extension of the Voronoi diagram, the power diagram introduces the “weight” characteristic to each site, that is, each site $\mathbf{s}_i$ is associated with a value w_i . Similar to the Voronoi diagram, the power diagram divides the domain Ω into n subregions $P(\mathbf{S}) = \{P(\mathbf{s}_i)\}_{i=1}^n$, and each subregion (also called “power cell”) $P(\mathbf{s}_i)$ of the site $\mathbf{s}_i$ is redefined:

$$P(\mathbf{s}_i) = \{\mathbf{s} \mid \|\mathbf{s} - \mathbf{s}_i\|^2 - w_i \leq \|\mathbf{s} - \mathbf{s}_j\|^2 - w_j, \forall j \neq i\} \quad (2)$$

Notably, the power diagram degenerates to a Voronoi diagram when all weights of sites are equal [2].

2.2 CCCPD

By imposing the centroidal constraint and capacity constraint to the ordinary power diagram, a CCCPD can be obtained. In other words, a CCCPD is a special power diagram, where 1) each site $\mathbf{s}_i$ is located in the mass center of its power cell $P(\mathbf{s}_i)$; and 2) the area/capacity of each $P(\mathbf{s}_i)$ is equal to the preset capacity constraint c_i of the site $\mathbf{s}_i$. The centroid and capacity of each power cell $P(\mathbf{s}_i)$ could be calculated:

$$\begin{cases} \mathbf{s}_i^* = \frac{\int_{P(\mathbf{s}_i)} \mathbf{s} \rho(\mathbf{s}) d\mathbf{s}}{\int_{P(\mathbf{s}_i)} \rho(\mathbf{s}) d\mathbf{s}} \\ m_i = \int_{P(\mathbf{s}_i)} \rho(\mathbf{s}) d\mathbf{s} \end{cases} \quad (3)$$

where $\rho(\mathbf{s})$ is the density of any point $\mathbf{s}$ located in the domain Ω . Aurenhammer et al. [3] point out that a CCCPD could be generated by minimizing the following term:

$$F(\mathbf{S}, \mathbf{W}) = \sum_{i=1}^n \int_{P(\mathbf{s}_i)} \rho(\mathbf{s}) \|\mathbf{s} - \mathbf{s}_i\|^2 d\mathbf{s} - \sum_{i=1}^n w_i (m_i - c_i) \quad (4)$$

Various studies have presented different algorithms to compute the power diagram. Balzer et al. [4] use the false-position method and the Lloyd’s method to generate the CCCPD, which is time-consuming. de Goes et al. [6] propose the Newton’s method to optimize the weights, and use the adaptive step-size gradient descent method to optimize the sites, which greatly improves the CCCPD computation efficiency. Xin et al. [24] further present the L-BFGS method with super-linear convergence to compute the CCCPD, but the construction of power diagram is still time-consuming. Recently, Zheng et al. [26] provide a GPU-accelerated method to improve the efficiency of power diagram construction rather than CGAL based method. Furthermore, they extend the capacity constraint in CCCPD to the general case, and introduce a hybrid capacity constrained centroidal power diagram (HCCCPD) [27]. In this paper, the power diagram computation framework in [6] is utilized to compute the image segmentation results.

2.3 Image abstraction

Existing methods for image abstraction mainly involves the segmentation based image abstraction method and the filter based image abstraction method. The former divides the input image into several subregions with similar features, each of which is filled with the same color to generate the abstract image. The Mean-Shift filter is used to separate the input image, and the abstract image is generated accordingly.

DeCarlo et al. [7] propose an interaction technique based on a visual perceptual model. The image abstraction relies on a hierarchical representation of visual forms in the input image, and the abstract images are formed by segmenting on different scales. However, an expensive eye tracker is required in this method, and the image abstraction is inefficient. Ren et al. [19] introduce the concept of superpixels, which are irregular blocks, consisting of pixels with similar features. By dividing the input image into a small number of superpixels rather than a large number of pixels, the abstract image could be generated, which greatly reduces the complexity of subsequent image processing. Achanta et al. [1] propose a Simple Linear Iterative Clustering (SLIC) method for segmenting the input image into superpixels. The SLIC method iteratively computes the clusters with the smallest distance, and these pixels with small distances are clustered in the same superpixel, thereby generating the abstract image. Rose et al. [20] propose a fuzzy clustering algorithm, where each pixel is assigned to clusters on basis of the probability of the distance between each pixel to clusters. According to idea in [20, 19], Gerstner et al. [10, 9] present an improved version of SLIC. A color palette is used to record the color of each superpixel and the abstract image is generated by simplifying the primal image colors. However, the segmentation based image abstraction method may generate the low-quality abstract image with unsmooth boundaries. In this paper, the image segmentation based on the power diagram is used to obtain the segmentation results, which produces the high-quality abstract image with smooth boundaries.

Wang et al. [22] first introduce the Mean-Shift filter to segment video data into continuous data, so as to achieve better image abstraction. Kang et al. [11] propose a stream-based variant of the bilateral filter to address the problems of abstract drawing and region smoothing. However, this method may lead to over-abstraction or under-abstraction regions. Winnemoller et al. [23] present an automatic, real-time image abstraction framework that uses a non-linear diffusion filter to smooth regions with low color contrast. Meanwhile, a Gaussian difference operator is utilized to process regions of high color contrast to achieve image abstraction. Kyprianidis et al. [16] present a framework for automatic non-realistic image processing techniques using anisotropic filtering methods. Furthermore, they introduce a Knwahara filtering algorithm with different anisotropy, but the obtained results are not satisfactory. Other researchers [25] provide some filter based image abstraction methods [21, 12], which may produce low-quality abstract images with artifacts and blurring. Besides, the degree of abstraction is difficult to control. In this paper, the number of sites and the density map provide our method with the ability of producing controllable abstract images.

3 Our method

In this section, we will introduce the pixelated image abstraction method via power diagram to generate high-quality abstract images.

3.1 Overall method

Given that the input image M is composed by a set of pixels $\{p_i\}_{i=1}^N$, where $N = w_M \times h_M$ is the number of pixels. Note that w_M is the width and h_M is the height of the input image. Each pixel p_i is associated with a color value $rgb(p_i)$ (RGB color is used in our work). The purpose of image abstraction is to produce an output image M' with a reduced color palette from the input image M , where the output image is of equal size to the input image. In this paper, we propose a novel pixelated image abstraction method via power diagram to generate high-quality abstract images with smooth boundaries and controllable abstraction degree.

The power diagram based pixelated image abstraction method in this paper mainly involves the following three modules:

- pre-processing for capturing the significant pixels and then calculating a density map;
- partitioning the input image into accurate polygonal subregions based on power diagram;
- coloring each polygonal subregion on basis of the average RGB color of the 3×3 grids centered at its centroid.

Consequently, the abstract images could be produced by merging color-filled polygonal subregions. The pseudo code of the pixelated image abstraction method via power diagram is provided in Algorithm 1, and the pipeline of the computational process is shown in Fig. 1.

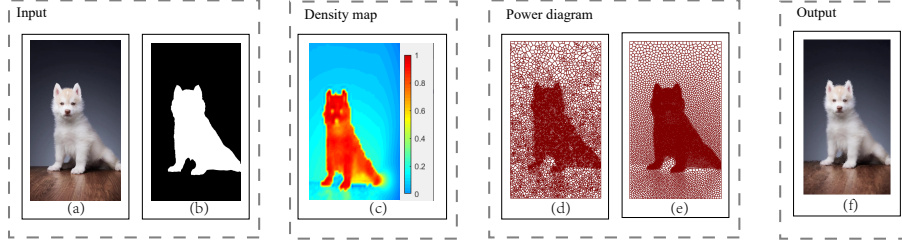


Fig. 1. The flow of the pixelated image abstraction method via power diagrams. (a) The input image; (b) image of the pre-processing; (c) the initialized density map; (d) the initialized power diagram with density adaptive; (e) the optimized power diagram, where each polygonal subregion with the same area constraint; and (f) the output image.

Algorithm 1: Pixelated image abstraction based on power diagram

Input: Origin image M , containing a set of pixels $\mathbf{p}_{in} = \{p_i\}_{i=1}^{w_M \times h_M}$, the number of sites n

Output: Abstract image M' (a set of pixels $\mathbf{p}_{out} = \{p_o\}_{i=1}^{w_M \times h_M}$)

- 1 *Pre-processing*: dividing the input image M into core and non-core parts
 - 2 Calculating the density map $\rho(\mathbf{s})$
 - 3 Generating n distinct sites $\mathbf{S} = \{\mathbf{s}_i\}_{i=1}^n$ on a domain of equal size to M , and setting each $w_i = 0$
 - 4 Setting the area/capacity constraint of each site $\mathbf{s}_i$ with average mass
 - 5 Optimizing the sites and weights to produce polygonal subregions with area constraint
 - 6 Computing the average color r_i of each site $\mathbf{s}_i$
 - 7 Filling each polygonal subregion $P(\mathbf{s}_i)$ with the color r_i
-

3.2 Pre-processing & Density initialization

Density map initialization is an essential step in our method, which affects the optimized polygonal subregions in the second module. Before initializing the density map, we should use a pre-processing step to capture these pixels with significant information in the input image, which greatly improves the quality of the output abstract image.

Pre-processing. Typically, a simple image consists of a foreground layer and a background layer. The former shows some significant information, such as the cat in Fig. 1, while the latter is the non-significant information, such as the table and wall in Fig. 1. In the image abstraction process, the pixels with significant information in the image should be preserved while these with non-significant information are ignored. With this regard, the pre-processing is utilized to divide the primal image into core and non-core parts with different levels of pixel importance. The pre-processing in our work could be directly implemented by existing segmentation techniques, such as OpenCV [5] and deep learning based

method [17]. The core part with high level importance is represented by M_α , and the remaining is denoted by M_β , where $M = M_\alpha \cup M_\beta$.

Density initialization. Density map plays an important role in our method. In what follows, we provide a general formulation for calculating density map of the input image based on the pre-processing results. We grayscale the input image, and then the grayscale value $g(p_i)$ of each pixel p_i is normalized to $col(p_i) \in [0, 1]$, where $p_i \in M$. To capture the importance of different pixels in the input image, the density value of each pixel p_i is calculated as follows:

$$\rho(p_i) = \begin{cases} a \cdot col(p_i) + b, & p_i \in M_\alpha \\ c \cdot col(p_i), & p_i \in M_\beta \end{cases} \quad (5)$$

where a , b and $c \in [0, 1]$ are the parameters for controlling the density map. Consequently, the density values of pixels with high-level importance (core, M_α) is larger than these with low-level importance (non-core, M_β) on basis of the input image.

3.3 Image segmentation based on the power diagram

Existing image segmentation based methods [1, 9] utilize “superpixels” to segment the input image into subregions, each of which is filled with the color of the corresponding superpixel. However, these methods may produce low-quality abstract images with unsmooth boundaries, such as these results in Fig. 8. Different from previous work, we use the power diagram to segment the input image into polygonal subregions, which are more suitable for capturing boundaries. Benefiting from this, our method could generate high-quality abstract images with smooth boundaries, which receives better visual effects to users. The polygonal division based on power diagram is introduced in the following.

Let Ω represent the domain of equal size to the input image M . $\mathbf{S} = \{\mathbf{s}_i\}_{i=1}^n$ denotes n distinct sites located in Ω , where each site $\mathbf{s}_i$ is associated with a weight w_i . The power diagram defines a partition of the domain Ω into n disjoint subregions $P(\mathbf{S}) = \{P(\mathbf{s}_i)\}_{i=1}^n$, as shown in Fig. 2(a). However, the subregions of the ordinary power diagram may be irregular, which contains many skinny power cells. Due to the possibility of producing many jagged power cells, the ordinary power diagram is useless for image abstraction.

In this paper, we use the centroidal power diagram with capacity constraint to produce the image segmentation results rather than the ordinary power diagram, as shown in Fig. 2(b). Specifically, we assign a capacity constraint c_i to each site $\mathbf{s}_i$, that is,

$$c_i = \frac{\int_{\Omega} \rho(\mathbf{s}) d\mathbf{s}}{n} \quad (6)$$

The area m_i of each polygonal subregion $P(\mathbf{s}_i)$ should satisfy: $m_i = c_i$. Meanwhile, the location of each site $\mathbf{s}_i$ should located in the centroid of the corresponding polygonal subregion $P(\mathbf{s}_i)$. To compute the CCCPD, researchers have provided various algorithms, including CPU-based computing algorithm [6, 24],

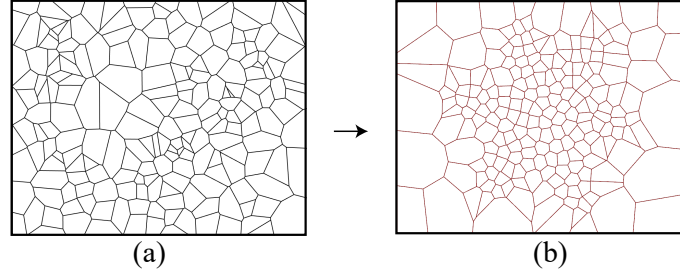


Fig. 2. Ordinary power diagram and centroidal power diagram with capacity constraints (CCCPD). (a) power diagram; and (b) CCCPD.

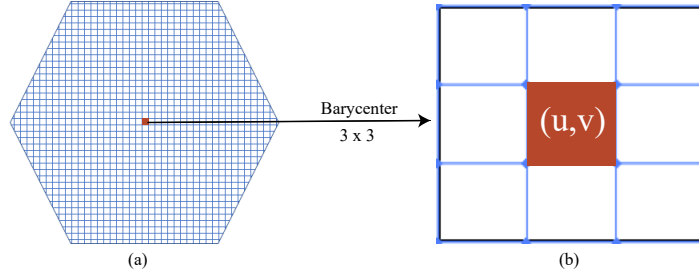


Fig. 3. The color filling of a power cell based on the average RGB color of the 3×3 grids centered at its barycenter. (a) optimized power cell; and (b) 3 grids centered at the barycenter of the power cell.

GPU-based computing algorithm[26], etc. In our work, we refer to previous work [6] and the CCCPD computing framework is used to compute the image segmentation based on power diagram.

Therefore, the image segmentation based on power diagram produces polygonal subregions with better characteristics that can capture the boundaries more accurately. Meanwhile, according to the initialized density map in previous text, the size of each polygonal subregion can be controlled. Specifically, the size of polygonal subregion is larger in low-level significant pixels while smaller in high-level significant pixels. Consequently, more important information (core) is preserved while less important information (non-core) is ignored during the image abstraction process of our method.

3.4 Color filling

After obtaining the optimized polygonal subregions from the image segmentation, each polygonal subregion should be filled with the corresponding RGB color. A straightforward way is to fill each polygonal subregion with the RGB color of its barycenter. However, this way may results in unsatisfactory abstract results due to the existing noisy pixels.

To eliminate the influence of noisy pixels, we use the average pixel color to fill each polygonal subregion $P(\mathbf{s}_i)$. First of all, we choose the 3×3 grids centered at the centroid $\mathbf{s}_i^* = (x_i, y_i)$ of each polygonal subregion $P(\mathbf{s}_i)$, as shown in Fig. 3. Let $pixel_{ij} = (u, v)$ denote the exact pixel coordinate in the primal image, where $u = \lfloor x_i \rfloor$ and $v = \lfloor y_i \rfloor$. The average RGB color of the 3×3 grids of $\mathbf{s}_i^*$ could be calculated as:

$$r_i = \frac{1}{9} \sum_{k=u-1}^{u+1} \sum_{t=v-1}^{v+1} rgb(k, t) \quad (7)$$

Accordingly, the pixels in each polygonal subregion $P(s_i)$ are filled with the average RGB color r_i , and the abstract image could be directly generated by filling all polygonal subregions.

4 Experiment

In this section, we present some experimental results to verify the effectiveness of our method. The proposed method was implemented in C++, and all experiments were performed on a computer with 3.6 GHz Intel (R) Core (TM) i7-9700K CPU and 16 GB memory. The default number of sites N is $5.0K$, and the default value of these parameters a , b and c are set to 0.5.

4.1 Results analysis

To verify the effectiveness of our method for image abstraction, we conduct an experiment on several different images. In this experiment, the number of sites n is set to $5.0K$, and the parameters for density map initialization is set to the default values. The results of pixelated image abstraction with power diagram are presented in Fig. 4. Benefiting from the density map initialization and the power diagram based image segmentation, our method could generate high-quality abstract images with smoother boundaries.

4.2 Abstract controlling

An fundamental characteristics of our method is the controllability of the abstraction degree. Both the number of sites and the density map could be taken into consideration to control the abstraction of images. Therefore, we perform two groups of experiments to demonstrate the image abstraction controllability of our method.

Number of sites controlling. Considering the influence of the number of sites on the generated results, we design the first group of experiments on three images, and the number of sites is selected from $1.0K$ to $10.0K$. Note that the parameters used to initialize the density map are set to the default values. The results of abstract images with different number of sites are provided in Fig. 5.

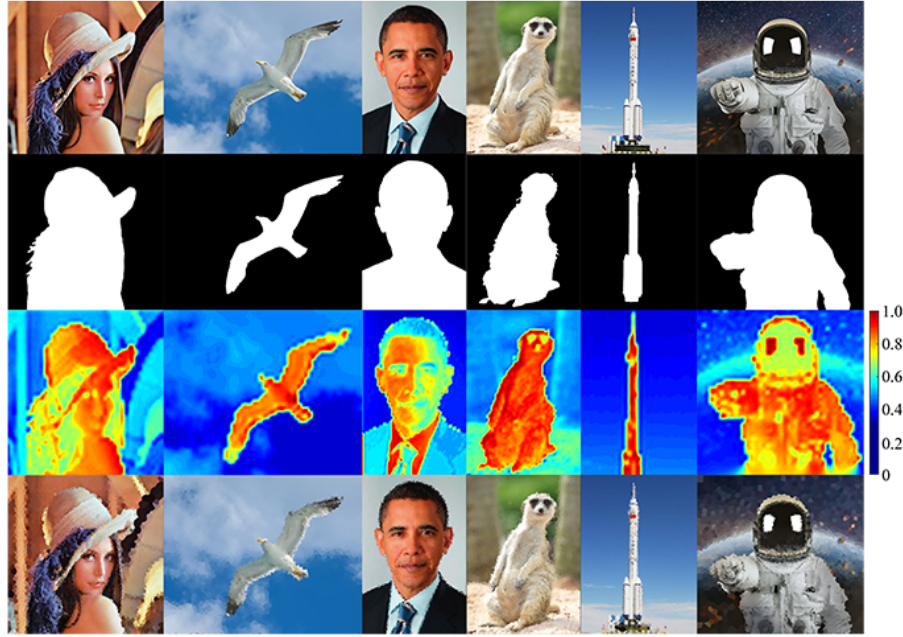


Fig. 4. Results of pixelated image abstraction with power diagram. From top to bottom: input images, pre-processing results, density maps, and output abstract images.

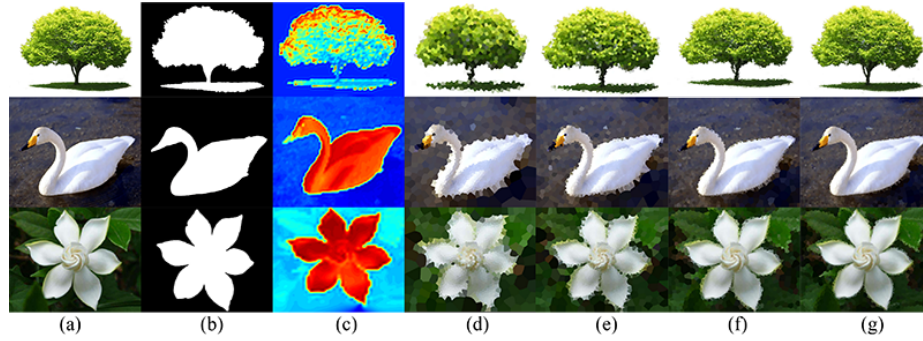


Fig. 5. The image abstraction results of our method with different number of sites. (a) Input images; (b) segmentation images; (c) density maps; (d) - (g) abstract images with 1.0K, 2.0K, 5.0K and 10.0K sites.

According to the results in Fig. 5, we can observe that the quality of abstract image improves with the gradually increasing of the number of sites. Specifically, when the number of sites is small (as shown in the fourth column in Fig. 5), our method produces the polygonal subregions with large sizes, which may cause unsmooth boundaries of the abstract images. As the number of sites increases,

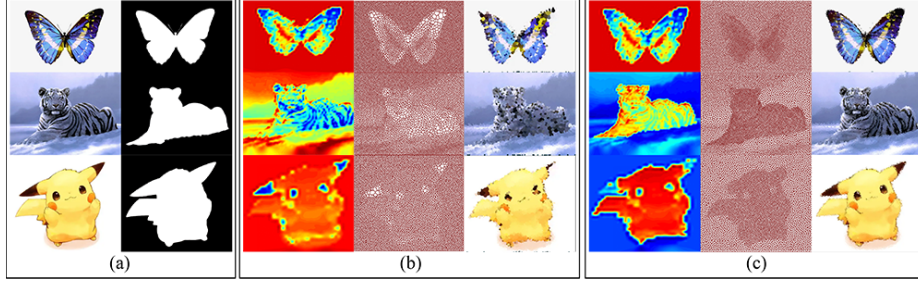


Fig. 6. The image abstraction results of density ablation experiments with our method. (a) Input images & pre-processing results; (b) results without density *Pair1-Exp1*; and (c) results with density *Pair1-Exp2*.

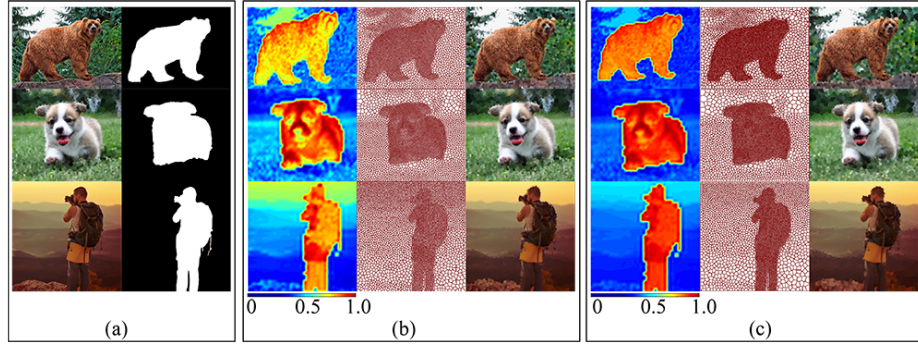


Fig. 7. The image abstraction results of our method with different densities. (a) Input images & pre-processing results; (b) results with density *Pair2-Exp1*; and (c) results with density *Pair2-Exp2*.

the sizes of polygonal subregions become smaller and the quality of generated abstract images improves.

Density map controlling. To further show the generated results with different densities, the second group of experiments is designed, where the number of sites is set to the default value, that is, $n = 5.0K$.

Density controlling 1. Firstly, the densities are selected as follows: 1) the densities are initialized without the consideration of pixels importance; and 2) the densities are initialized based on the pixels importance. Specifically, the density of each pixel in this experiment is calculated as follow:

$$\begin{aligned}
 \text{Pair1-Exp1} : \rho(p_i) &= \text{col}(p_i), \quad p_i \in M \\
 \text{Pair1-Exp2} : \rho(p_i) &= \begin{cases} 0.5 \cdot \text{col}(p_i) + 0.5, & p_i \in M_\alpha \\ 0.5 \cdot \text{col}(p_i), & p_i \in M_\beta \end{cases} \quad (8)
 \end{aligned}$$

The experimental results of the first pair experiments are provided in Fig. 6.

Density controlling 2. Then, a new pair of experiments is performed, and the importance of pixels is taken into account. To be specific, the densities of pixels with high-level importance are larger than those with low-level importance. However, the difference lies in the density variability between core and non-core parts, and the density map calculation is as follow:

$$\begin{aligned} \text{Pair2-Exp1} : \rho(p_i) &= \begin{cases} 0.5 \cdot \text{col}(p_i) + 0.5, & p_i \in M_\alpha \\ 0.5 \cdot \text{col}(p_i), & p_i \in M_\beta \end{cases} \\ \text{Pair2-Exp2} : \rho(p_i) &= \begin{cases} 0.3 \cdot \text{col}(p_i) + 0.7, & p_i \in M_\alpha \\ 0.3 \cdot \text{col}(p_i), & p_i \in M_\beta \end{cases} \end{aligned} \quad (9)$$

The abstract image results are presented in Fig. 7.

According to the results in Fig. 6 and Fig. 7, we can observe that our method achieves controllable abstract images with different densities. Specifically, when the importance of pixels is not considered (as shown in Fig. 6(b)), many detailed features in the abstract images are lost, such as the textures on the tiger. On the contrary, the results in Fig. 6(c) demonstrate the validity of the density map in our method, which takes into account the importance of pixels. Thus, the generated results retain more features of the primal images. Considering the density variations of core and non-core parts, the results in Fig. 7 show that higher densities could retain more detailed information in the core part, while lower densities could ignore the non-important information in the primal image. More generally, the density map initialization in 5 could be extended to more general cases, such as Gaussian or non-linear densities.

4.3 Comparison

To further illustrate the effectiveness of our method with power diagram, we compare against the improved SLIC method of Gerstner et al.[9] as a representative of state-of-the-art image abstraction methods based on image segmentation. The improved SLIC method [9] divides the input image into superpixels and uses a simple linear iterative clustering method to optimize the superpixels. In this experiment, we select six different images as inputs of our method and the improved SLIC method, and the generated abstract images of different methods are presented in Fig. 8. To maintain the consistency, the number of superpixels in the improved SLIC method is set to 4800, that is same to the number of sites in our method. Notably, the parameters used for density map initialization in our method are set to the default values.

From the results in Fig. 8, we can observe that the method proposed in this paper outperforms the improved SLIC method. Specifically, compared with the improved SLIC method [9], the abstract images generated by our method retains more features, i.e., the eyes of the rabbit in the second column and the ears of the cats in the last column in Fig. 8. Besides, in terms of the boundaries of the generated abstract images, our method produces higher quality results

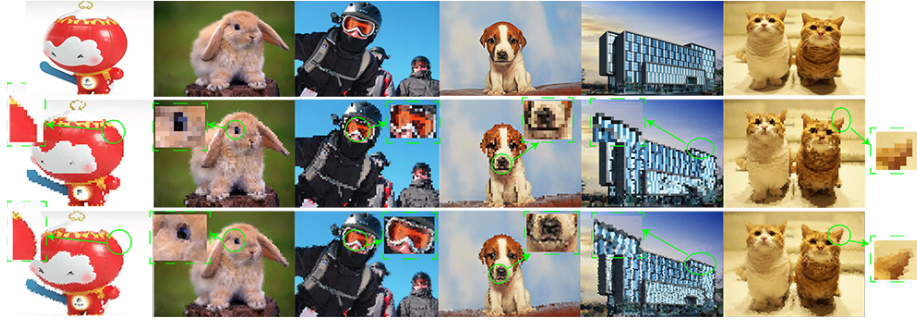


Fig. 8. Comparison results of different image abstraction methods. From top to bottom: input images, results of the improved SLIC method [1], and results of our method.

with smooth boundaries, while the improved SLIC method obtains results with unsmooth boundaries.

5 Conclusion

We propose a novel image abstraction method based on power diagram to generate high-quality abstract images with smooth boundaries and controllable abstraction degree. Based on the power diagram, the domain with equal size to the input image is partitioned into several polygonal power cells, which is then optimized to meet the equal capacity constraints. The abstract image is generated by coloring each pixel in the power cell with the average color of 3×3 grids centered at the barycenter of its power cell. Compared with other segmentation based image abstraction method, the proposed method use the power diagram to partition the primal domain into polygonal cells, which is more suitable for capturing the boundaries. Besides, both the density and the number of sites could dominate the degree of image abstraction.

Limitations and future work. Despite the density map of equal size to the input image is built based on the saliency segmentation image, the generated image is not satisfactory for complex images. In our future work, we will extend the current work to make it more applicable for complex image abstraction.

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